Symbolic Mechanism Discovery via Best-First Search in Mathematical Operation Space

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Abstract

We introduce a framework for automated discovery of interpretable predictive formulas from labeled scientific data using best-first search through a space of mathematical operations. Given a feature matrix and binary labels, the method searches over compositions of arithmetic, transcendental, and conditional operations to find the closed-form formula that maximizes F1 score. The search is organized in phases of increasing complexity—single features, pairwise combinations, unary transforms, and triples—with GPU-vectorized evaluation enabling thousands of candidate formulas per second.

The framework provides two outputs: (i) the best interpretable formula at each depth, and (ii) the *formula ceiling*—the maximum F1 achievable by any closed-form expression of bounded complexity. Comparing this ceiling to a black-box ensemble quantifies the *irreducible complexity gap*: how much predictive structure resists symbolic expression.

Applied to predicting periodic orbits in the gravitational three-body problem (15,000 configurations, 23 Hessian-derived features, GPU-integrated ground truth), the method discovers that periodicity is predicted by the norm $\sqrt{n_{\text{clusters}}^2 + \lambda_{\min}^2}$ (F1 = 0.48). The ceiling is robust across all tested operations. A gradient-boosted ensemble reaches F1 = 0.86, yielding an irreducible complexity gap of 0.38—meaning 38% of the predictive structure in three-body dynamics cannot be expressed as a depth-4 formula.

Code and data: `https://github.com/ahb-sjsu/astar-symbolic-discovery`.

1 Introduction

A central goal of computational science is to discover interpretable laws from data. This problem has been addressed by genetic symbolic regression [1, 4], neural equation discovery [2], and classical AI systems [3]. Genetic programming explores large formula spaces but provides no optimality guarantee at a given depth and often produces bloated expressions. Neural methods discover accurate but opaque representations. Classical systems require domain-specific search heuristics.

We propose a structured search through mathematical operation space using *best-first search* with F1 score as the ranking criterion. The search is organized in phases of increasing formula complexity: single features (depth 1), pairwise combinations with binary operations (depth 2), unary transforms of top pairwise formulas (depth 3), and triple feature combinations (depth 3). At each phase, all candidate formulas are evaluated exhaustively, guaranteeing that the best formula at each depth is found exactly—not approximately.

This phased exhaustive approach avoids the admissibility issues of heuristic A* search on formula spaces (where defining an admissible heuristic for future F1 improvement is nontrivial) while preserving the key practical benefit: the best formula at each depth is identified with certainty.

Our framework provides two outputs that existing methods do not:

1. The **best interpretable formula** at each complexity level, found by exhaustive search within that level.

2. The **formula ceiling**—the maximum F1 achievable by any closed-form expression of bounded depth. Comparing this ceiling to a black-box model quantifies the *irreducible complexity gap*, telling the practitioner when further symbolic search is futile.

2 Method

2.1 Problem Formulation

Definition 1 (Symbolic Prediction Problem)**.** *Given a feature matrix $X \in \mathbb{R}^{N \times d}$ and binary labels $y \in \{0,1\}^N$, find a formula $f : \mathbb{R}^d \to \mathbb{R}$ and threshold $\tau \in \mathbb{R}$ that maximize $F_1(\hat{y}, y)$ where $\hat{y}_i = \mathbf{1}[f(X_i) > \tau]$, subject to* $\mathrm{depth}(f) \leq D$.

2.2 Operation Registry

The search space is defined by three operation classes:

Leaf nodes (depth 0): Each of the d features x_j.

Unary operations (depth +1): $\{\log|\cdot|, \sqrt{|\cdot|}, (\cdot)^2, 1/(\cdot), -(\cdot), |\cdot|, \sigma(\cdot), \tanh(\cdot)\}$ (8 operations).

Binary operations (depth +1): $\{+, -, \times, \div, \max, \min, \mathrm{hypot}, (\cdot - \cdot)^2, \text{harmonic-mean}, \text{geometric-mean}\}$ (10 operations).

2.3 Phased Exhaustive Search

Rather than a single tree search, we organize the exploration into four phases (Algorithm 1). Each phase exhaustively evaluates all formulas at its complexity level.

Threshold sweep. For each candidate formula f, we compute $v_i = f(X_i)$ for all N samples, then evaluate F1 at percentile thresholds $\{20, 30, 40, \ldots, 95\}$ of the value distribution. Both $v > \tau$ and $v < \tau$ predictors are tested. The best threshold is retained. This sweep is vectorized and adds negligible cost.

GPU acceleration. Features are stored as GPU arrays (CuPy). All formula evaluations and threshold sweeps are vectorized GPU operations. Phase 2 evaluates $d^2 \times |\mathcal{B}|$ formulas (e.g., $23^2 \times 10 = 5{,}290$) in seconds.

Completeness within each phase. Phases 1–2 are exhaustive over all feature (pairs) and operations. Phase 3 applies all unary transforms to the top-K results from Phase 2. Phase 4 tests multiple combination patterns over the top-K single features. The search is complete at each phase for the specified formula structure; no formulas of that structure are missed.

2.4 Scalability

The formula count per phase is: Phase 1: $O(d)$; Phase 2: $O(d^2 \cdot |\mathcal{B}|)$; Phase 3: $O(K \cdot |\mathcal{U}|)$; Phase 4: $O(\binom{K}{3} \cdot P)$ where P is the number of triple-combination patterns.

For $d = 23$, $|\mathcal{B}| = 10$, $|\mathcal{U}| = 8$, $K = 50$, $P = 9$, the total is approximately 6,200 formulas—evaluated in under one minute on a single GPU. Depth 5–6 formulas (Phase 2 composed with Phase 2) would require $O(d^4 \cdot |\mathcal{B}|^2) \approx 28$ million formulas, which is tractable with GPU vectorization but would take hours. We do not explore beyond depth 4 in this work.

Algorithm 1 Phased Exhaustive Symbolic Search

Require: Features $X \in \mathbb{R}^{N \times d}$, labels y, operations $\mathcal{U}$ (unary), $\mathcal{B}$ (binary)
Ensure: Best formula and F1 at each depth

Phase 1: Single features (depth 1)
for $j = 1, \ldots, d$ **do**
  Evaluate $F_1^*(x_j)$ via threshold sweep
$R_1 \leftarrow$ top-K features by F1

Phase 2: Pairwise combinations (depth 2)
for $x_i, x_j \in \{1, \ldots, d\}$, $b \in \mathcal{B}$ **do**
  Evaluate $F_1^*(b(x_i, x_j))$ via threshold sweep
$R_2 \leftarrow$ top-K pairwise formulas

Phase 3: Unary transforms (depth 3)
for $f \in R_2$, $u \in \mathcal{U}$ **do**
  Evaluate $F_1^*(u(f))$ via threshold sweep
$R_3 \leftarrow$ top-K depth-3 formulas

Phase 4: Triple combinations (depth 3)
for $x_i, x_j, x_k \in R_1$, pattern p **do**
  Evaluate $F_1^*(p(x_i, x_j, x_k))$ via threshold sweep
return best formula across all phases

2.5 Formula Ceiling and Irreducible Complexity Gap

Definition 2 (Formula Ceiling)**.** *The formula ceiling C_D at depth D is the maximum F1 achieved by any formula of depth $\leq D$ found across all phases: $C_D = \max_{phases} F_1^*$.*

Definition 3 (Irreducible Complexity Gap)**.** *Given a black-box model with F1 score F_1^{bb} trained on the same features, the irreducible complexity gap at depth D is $G_D = F_1^{\text{bb}} - C_D$.*

$G_D > 0$ means the data contains predictive structure that cannot be captured by depth-D formulas. G_D is a property of the data and the feature set, not of the search algorithm. If C_D converges (does not increase with additional operations or deeper phases), further symbolic search is futile.

Proposition 1. *C_D is non-decreasing in D and bounded above by F_1^{Bayes}. If $C_{D+1} = C_D$ for a sufficiently rich operation set, the ceiling has converged.*

3 Case Study: Three-Body Periodic Orbit Prediction

3.1 Setup

We compute 23 features from the Hessian of the gravitational three-body Hamiltonian at 15,000 random configurations of equal-mass bodies ($m_1 = m_2 = m_3 = 1$) in Jacobi coordinates. Features include: numerical matrix rank (threshold $\epsilon = 10^{-6}$), potential block eigenvalues and their ratios, frequency cluster counts, spectral entropy, Hamiltonian energy, body separation ratios, and derived quantities. Ground truth is obtained by symplectic leapfrog integration ($\Delta t = 0.005$, 8×10^4 steps) on an NVIDIA GV100 GPU, with periodicity defined as phase-space return to within 5% of initial extent. Batch size is maximized automatically using the batch-probe library [5].

3.2 Search Results

Table 1 summarizes the results across all four phases.

Table 1: Formula search results by phase (15,000 configs, 23 features).

Phase	Best formula	F1	Formulas tested
1 (single)	`n_clusters`	0.35	23
2 (pairwise)	`n_clust +` $\lambda_{\min}$	0.48	5,060
3 (unary)	`hypot(n_clust,` $\lambda_{\min}$`)`	0.48	400
4 (triple)	(none above 0.30)	—	504
Ceiling		**0.48**	**5,987**

The ceiling converges at Phase 2: unary transforms (Phase 3) and triple combinations (Phase 4) provide no improvement. All 20 top formulas are algebraically equivalent to $\alpha \cdot n_{\text{clusters}} + \beta \cdot \lambda_{\min}$ for various α, β. The Euclidean norm variant

$$\hat{y} = \mathbf{1}\left[\sqrt{n_{\text{clusters}}^2 + \lambda_{\min}^2} > \tau\right] \tag{1}$$

is representative of the equivalence class.

3.3 Irreducible Complexity Gap

Table 2: Formula ceiling vs. black-box models (51,000 configs, 3 mass ratios).

Model	F1	Gap from ceiling
Random baseline (5% rate)	0.07	—
Separation ratio $r_2/r_1 > 10$	0.29	—
Best depth-4 formula (Eq. 1)	0.48	0.00 (= ceiling)
Decision tree (depth 5)	0.58	−0.10
Gradient boosting (100 trees)	0.86	−0.38

The irreducible complexity gap is $G_4 = 0.86 - 0.48 = 0.38$. The decision tree (F1 = 0.58) captures some of the gap via piecewise-constant splits; the remaining 0.28 requires the nonlinear interactions of an ensemble. This means that 38% of the predictive structure in three-body periodicity resists expression as any depth-4 formula over Hessian features.

3.4 Physical Interpretation

The discovered formula combines two independent necessary conditions:

- $n_{\text{clusters}} \geq 2$: the eigenvalue frequency spectrum separates into fast (inner binary) and slow (outer body) groups, indicating timescale separation.
- $\lambda_{\min} > 0$: the smallest Hessian eigenvalue is nonzero, indicating the system is gravitationally bound (not escaping).

The Euclidean norm indicates these contribute independently—there is no synergy or interaction between them. Periodicity requires *both* separated timescales *and* sufficient binding.

3.5 Comparison to PySR

To contextualize our method, we note how it differs from PySR/genetic symbolic regression [4] on this task:

Completeness vs. exploration. Our phased search evaluates all $d^2 \times |\mathcal{B}|$ pairwise formulas exhaustively. Genetic methods sample this space stochastically and may miss the global optimum at a given depth, but can explore deeper formula trees (depth 10+) that our exhaustive approach cannot reach.

Formula ceiling. Our framework explicitly computes the ceiling and the gap to a black-box model. PySR produces formulas of increasing complexity but does not systematically measure when further complexity stops helping.

Speed. Our GPU-vectorized evaluation of ∼6,000 formulas completes in under one minute. PySR typically requires minutes to hours depending on population size and generations. For depth ≤ 4 with moderate feature counts ($d < 30$), exhaustive search is faster; for deeper or wider searches, genetic methods are more practical.

A direct head-to-head comparison on standard symbolic regression benchmarks (e.g., AI Feynman [2]) is left for future work; our current results demonstrate the method on a novel scientific prediction task rather than on benchmark recovery.

4 Discussion

4.1 When to Use This Method

The phased exhaustive approach is most valuable when: (i) the feature count is moderate ($d < 50$, so d^2 is tractable); (ii) the practitioner wants a *guaranteed* best formula at each depth, not a stochastic approximation; (iii) the formula ceiling and irreducible complexity gap are of interest (e.g., to decide whether a problem admits a simple law or requires a complex model).

For large feature spaces ($d > 100$) or deep formula trees (depth > 5), genetic methods or neural symbolic regression are more appropriate.

4.2 The Ceiling as a Scientific Diagnostic

The formula ceiling is not just a performance metric—it is a statement about the data. A converged ceiling means the relationship between features and labels has a *fundamental* nonlinearity that no short formula can capture. In our three-body case study, the ceiling of 0.48 at depth 4 with the gap of 0.38 to gradient boosting tells us that three-body periodic orbit prediction is genuinely complex: it depends on nonlinear feature interactions that the underlying physics does not reduce to a compact law at the level of Hessian features.

This diagnostic is novel. Existing symbolic regression methods report the best formula found but do not quantify *how much better* a more complex model could do. The ceiling provides a principled stopping criterion: if $C_{D+1} = C_D$ and $G_D > 0$, stop searching for formulas and use an ensemble.

4.3 Limitations

The search is exhaustive only within each phase's structure. Phase 2 (pairwise) is complete; Phase 3 applies unary transforms only to top-K pairwise results, not all; Phase 4 tests only pre-specified triple patterns. A formula with structure not matching any phase template (e.g., $\log(x_1 + x_2) \cdot x_3/x_4$) will be missed.

The method does not discover features—it evaluates pre-computed ones. The quality of the discovered formula depends entirely on the feature engineering.

The F1 metric weights precision and recall equally. Domain-specific objectives (e.g., prioritizing recall for safety applications) could change the optimal formula.

5 Conclusion

We have introduced a phased exhaustive search framework for symbolic mechanism discovery that provides two novel contributions: guaranteed-best formulas at each complexity level, and the formula ceiling as a diagnostic for irreducible complexity. Applied to the three-body problem, the method discovers that periodicity depends on two Hessian features (frequency cluster count and minimum eigenvalue) with a hard ceiling at F1 ≈ 0.48, while ensemble methods reach 0.86. The 38% irreducible complexity gap quantifies the fundamental nonlinearity of three-body dynamics at the level of Hessian features.

The method is domain-general, requiring only a feature matrix and binary labels. It is practical for feature spaces up to $d \approx 50$ and formula depths up to 4, with GPU acceleration enabling evaluation of $\sim$6,000 formulas in under one minute.

References

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